Project 1 Report

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Problem 1 — Arithmetic & Logical Calculations

The following expressions were evaluated in R:

Expression	Result
`123 * 453`	`55719`
`5^2 * 40`	`1000`
`TRUE & FALSE`	`FALSE`
`TRUE | FALSE`	`TRUE`
`75 %% 10`	`5`
`75 / 10`	`7.5`

Problem 9 — Comments on second_vector Operations

Each line of code is explained with a one-sentence comment:

```
second_vector + 20
```

Adds 20 to every element in second_vector, shifting all values up by 20.

```
second_vector * 20
```

Multiplies every element in second_vector by 20, scaling all values proportionally.

```
second_vector >= 20
```

Checks which elements in second_vector are greater than or equal to 20, returning a logical TRUE/FALSE vector.

```
second_vector != 20
```

Checks which elements in second_vector are not equal to 20, returning a logical TRUE/FALSE vector.

Problem 18 — Boolean Bracket Extraction

The code `first_vector[c(FALSE, TRUE, FALSE, TRUE)]` uses a logical vector to filter elements. Only positions marked TRUE are returned — the 2nd element (12) and the 4th element (5). Positions marked FALSE are excluded.

Result: `[1] 12 5`

Problem 19 — Logical Comparison on second_vector

The code `second_vector >= 10` returns a logical TRUE/FALSE vector indicating which elements of second_vector have a value greater than or equal to 10. It does not filter the vector — only the condition is evaluated for each element.

Problem 20 — Filtering with Bracket Notation

The code `one_to_one_hundred[one_to_one_hundred >= 20]` places a logical condition inside bracket notation to filter the vector. The condition produces a TRUE/FALSE vector used as an index to return only elements with values of 20 or greater — resulting in numbers 20 through 100.

Problem 33 — Summary of ds_salaries Dataset

Output of `summary(first_dataframe)` in R:

```
       X            work_year      experience_level   employment_type
Min.   :  0.0    Min.   :2020    Length:607         Length:607
1st Qu.:151.5    1st Qu.:2021    Class :character   Class :character
Median :303.0    Median :2022    Mode  :character   Mode  :character
Mean   :303.0    Mean   :2021
3rd Qu.:454.5    3rd Qu.:2022
Max.   :606.0    Max.   :2022
```

```
  job_title          salary          salary_currency    salary_in_usd
Length:607         Min.   :   4000   Length:607         Min.   :  2859
Class :character   1st Qu.:  70000   Class :character   1st Qu.: 62726
Mode  :character   Median : 115000   Mode  :character   Median :101570
                   Mean   : 324000                      Mean   :112298
                   3rd Qu.: 165000                      3rd Qu.:150000
                   Max.   :30400000                     Max.   :600000
```

```
employee_residence   remote_ratio     company_location   company_size
Length:607           Min.   :  0.00   Length:607         Length:607
Class :character     1st Qu.: 50.00   Class :character   Class :character
Mode  :character     Median :100.00   Mode  :character   Mode  :character
                     Mean   : 70.92
                     3rd Qu.:100.00
                     Max.   :100.00
```